Chapter 3

Censored Observations and Truncated Variables

The need to formulate probability structures for censored observations or truncated variables can arise in a variety of ways. Two of the most common will be briefly described in this section. When continuous variables have been associated with observable quantities, but values beyond some point in the support (either to the right or to the left) are not observed exactly, those observations are censored. When we restrict the support of the distribution of random variables to avoid the possibility of modeling impossible realizations (e.g., negative values for something that can only be positive) we do so by truncating the distribution. There are a number of types of censoring that can occur and there are two main types of truncation that get used. We will discuss these in turn.

3.1 Censored Random Variables

Probably the most classic examples of censoring occurs in studies connected with reliability or survival analysis in which response random variables are associated with time to some event. Examples include time to failure for machine components or time to death for cancer patients. As already mentioned, there are a number of types of censoring that can occur. Divisions used to categorize types of censoring include Right, Left and Interval censoring, and Type I versus Type II censoring. The types of censoring that may arise in a study are largely determined by study design. The distinction between what are called Type I and Type II censoring is whether the censoring point is fixed in a study or varies across sampling units, perhaps due to some random mechanism. Unless we are trying to model a censoring mechanism along with a distribution of time-to-event, the difference between Type I and Type II censoring does not impact the analysis, and we will confine our discussion to right, left, and interval censoring.

3.1.1 Types of Censoring

1. Interval Censoring.

   Interval censoring occurs when a random variable Y_i is reported to have assumed value y_i, where all that is known is that the actual value of Y_i is in the interval $(y_i - \Delta_{1,i},\, y_i + \Delta_{2,i})$. Usually, $\Delta_{1,i} = \Delta_{2,i} = \Delta_i$ and often $\Delta_i = \Delta$ for all i in a set of random variables. In truth, all data are interval censored since even quantities modeled with continuous random variables are reported to some finite precision. Considering all data as interval censored is the easiest way to see that censoring does not necessarily have to occur with respect to variables that represent

time-to-event. Within that context, however, interval censoring can arise either from precision of data records or as a result of times of observation. Consider an acute toxicity test in which organisms are exposed to a potential toxicant, and time to death is to be recorded for each organism. The organisms may only be observed every 4 hours, or maybe even only once a day. This produces interval censoring as a result of study design or observational protocol.

2. Right Censoring.

   Right censoring is probably the most common type of censoring encountered in practice and occurs when some portion of a set of random variables are known only to have values greater than some *censoring times*. Consider a set of random variables $\{Y_i : i = 1, \ldots, n\}$ that are associated with time to some event for sampling units $i = 1, \ldots, n$. What is recorded is either the observation that Y_i has assumed the value y_i or the observation that $Y_i > c_i$ for some known values c_i; $i = 1, \ldots, n$. It is assumed that the censored observations can also be identified and we can let $\delta_i = 1$ if Y_i leads to a censored observation, and $\delta_i = 0$ if the observation of Y_i is not censored. What is recorded in the available data then are pairs of values $\{(min\{y_i, c_i\}, \delta_i) : i = 1, \ldots, n\}$. There is really no standard convention for reporting censored data, however, and one needs to exercise caution to determine what is contained in a given set of data. For example, some studies might be reported in the format $\{(min\{y_i, c_i\}, \delta_i) : i = 1, \ldots, n\}$ but with $\delta_i = 0$ indicating censoring (rather than $\delta_i = 1$).

   The censoring times may all be the same, which is the case in a study that is started at some fixed time that can be set to 0, and observation

is ended at some other fixed time T. This might occur, for example, if n automobile batteries are all connected to a device that produces constant discharge, and the time until battery death is recorded, but only up to 48 hours at which time the apparatus is dismantled. In other studies, the censoring times might vary, such as when patients drop out of a clinical trial.

3. Left Censoring.

   Although less common that right censoring, left censoring occurs when some portion of a set of random variables are known only to have values less than some set of censoring times. This is just the mirror image of right censoring, and data that are recorded have the form of pairs $\{(max\{y_i, c_i\}, \delta_i) : i = 1, \ldots, n\}$, where the δ_i are again indicators for observations that are censored. The most common situation I know of in which left censoring occurs is in chemical analysis in which there is a *limit of detection* below which the equipment being used cannot detect a compound. This occurs, for example, in the analysis of environmental samples (e.g., soil or water) for toxic substances such as PCBs (polychlorinated biphenyls) or PAHs (polycyclic aromatic hydrocarbons).

3.1.2 A Major Assumption

One way to think about censored observations is that they constitute missing values or involve missing information, and a crucial assumption about problems involving censored data is essentially the same as a crucial assumption about problems involving missing data. That assumption is that the censoring mechanism is independent of the mechanism producing observed values, which is similar to an assumption of missing at random or, in a situation

lacking covariates, missing completely at random.

The importance of this assumption is obvious if one is attempting to construct a model to represent a mechanism of time-to-event. Consider, for example, a clinical trial for a new treatment for althlete's foot (a fungal infection that usually occurs on the bottom of the feet and between the toes, but can spread to other parts of the body). The observable response might be time to complete cure. If people drop out of the trial because they believe the treatment has not been successful for them, and go back to using their old remedy, then the results of the study have been compromised (at least in terms of the type of analysis discussed here).

3.1.3 Likelihoods for Censored Variables

Regardless of the particular model used or whether estimation and inference are to proceed using likelihood or Bayesian techniques, we need the likelihood for any model containing censored observations. We will assume that the assumption of independence between censoring and time-to-event mechanisms is justified and that the observation of sampling units justifies an assumption of independence among random variables that are associated with observations on those units. Consider, then, a setting in which we have continuous random variables $\{Y_i : i = 1, \ldots, n\}$ associated with times to some event for sampling units, and we are willing to model these variables as independent and identically distributed with common probability density function $f(y_i|\boldsymbol{\theta})$ having support Ω and some unknown parameter $\boldsymbol{\theta}$. Recorded observations of these random variables may be interval, right, or left censored Our objective is to determine a likelihood for estimation of $\boldsymbol{\theta}$ using either likelihood or Bayesian methods.

It is most convenient here to develop likelihoods based entirely on probabilities, rather than densities, and this can be motivated by the recognition that all recorded data are actually censored in some manner, as mentioned previously. Let $\mathcal{I} = \{i : \text{ observation } i \text{ is interval censored}\}$. Let $\mathcal{R} = \{i : \text{ observation } i \text{ is right censored}\}$. Let $\mathcal{L} = \{i : \text{ observation } i \text{ is left censored}\}$. Rather than using the terms *right censoring time* and *left censoring time* we will consider *lower possible value* and *upper possible value* for each observation. Let $\{c_{i,L} : i = 1, \ldots, n\}$ and $\{c_{i,U} : i = 1, \ldots, n\}$ denote these values, respectively. Then each observation, regardless of whether it is left, right, or interval censored, must be such that the corresponding random variable satisfies $c_{i,L} < Y_i < c_{i,U}$. If an observation is left censored, we take $c_{i,L} = -\infty$ and if an observation is right censored we take $c_{i,U} = \infty$. In a data set for which we will consider all observations censored, we now only need values for $c_{i,L}$, $c_{i,U}$ and censoring type (with values $\mathcal{L}$, $\mathcal{R}$, or $\mathcal{I}$).

The likelihood for $\boldsymbol{\theta}$ based on observations $\{y_i : i = 1, \ldots, n\}$ may now be written as

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} L_i(\boldsymbol{\theta}) = \prod_{i=1}^{n} Pr(c_{i,L} \leq Y_i \leq c_{i,U}). \tag{3.1}$$

In (3.1),

$$Pr(c_{i,L} \leq Y_i \leq c_{i,U}) = \begin{cases} Pr(-\infty \leq Y_i \leq c_{i,U}) = F(c_{i,U}|\boldsymbol{\theta}) & \text{if } i \in \mathcal{L} \\ Pr(c_{i,L} \leq Y_i \leq \infty) = 1 - F(c_{i,L}|\boldsymbol{\theta}) & \text{if } i \in \mathcal{R} \\ Pr(c_{i,L} \leq Y_i \leq c_{i,U}) = F(c_{i,U}|\boldsymbol{\theta}) - F(c_{i,L}|\boldsymbol{\theta}) & \text{if } i \in \mathcal{I}. \end{cases} \tag{3.2}$$

Often, observations that are not either right or left censored will be taken as exact values rather than interval censored, as we usually do for observations corresponding to continuous random variables when we are not considering censoring. The likelihood will now be written as a combination

of probabilities (for censored observations) and densities (for exact observations).

$$L(\boldsymbol{\theta}) = \prod_{i \in \mathcal{L}} F(c_{i,U}|\boldsymbol{\theta}) \prod_{i \in \mathcal{R}} [1 - F(c_{i,L}|\boldsymbol{\theta})] \prod_{i \in \mathcal{I}} f(y_i|\theta) \tag{3.3}$$

This amounts, essentially, to replacing $F(c_{i,U}|\boldsymbol{\theta}) - F(c_{i,L}|\boldsymbol{\theta})$ in the last line of (3.2) with the density value $f(y_i|\boldsymbol{\theta})$. If all data are recorded to the same precision, then the intermediate value theorem from integral calculus holds, and

$$\begin{aligned} F(c_{i,U}|\boldsymbol{\theta}) - F(c_{i,L}|\boldsymbol{\theta}) &= F(y_i + \Delta|\boldsymbol{\theta}) - F(y_i - \Delta|\boldsymbol{\theta}) \\ &= \int_{y_i-\Delta}^{y_i+\Delta} f(t|\boldsymbol{\theta})\, dt \\ &\approx 2\Delta f(y_i|\boldsymbol{\theta}). \end{aligned}$$

The result is that the likelihood obtained by using density values for observations assumed to be exact is proportional to the likelihood given in (3.1) and (3.2) and thus has the same shape and mode.

Example 3.1 – Remission in Leukemia Patients

Lawless (1982, p. 136) presents data on length of a remission period in leukemia patients given a particular therapy. After therapy is given there is a period of remission for each patient that we will model as random. In particular, let $\{Y_i : i = 1, \ldots, n\}$ be random variables associated with the time to the end of the remission period for n patients. Of course, in a study like this not all patients were given the therapy at the same time, so when the study ended some patients had been under observation longer than some other patients. In addition, at the time data were collected some patients were still in remission and are thus right censored. The data were recorded in units of week. Here, there are only two censoring classes, $\mathcal{I}$ for patients who

came out of remission during the study and $\mathcal{R}$ for patients still in remission at the end of the study. For $Y_i \in \mathcal{I}$ we will take $c_{i,L} = y_i - 1/2$ and $c_{i,U} = y_i + 1/2$. For $Y_i \in \mathcal{R}$ the recorded value is $c_{i,L}$. The data are given in Table 3.1 in which patient index has been ordered with increasing recorded value. In a set of course notes for Statistics 415, Meeker (2003) fits both exponential and lognormal models to these data.

Patient	1	2	3	4	5	6	7	8	9	10
Recorded Value	1	1	2	2	2	6	6	6	7	8
$c_{i,L}$	0.5	0.5	1.5	1.5	1.5	5.5	5.5	5.5	6.5	7.5
$c_{i,U}$	1.5	1.5	2.5	2.5	2.5	6.5	6.5	6.5	7.5	8.5
Patient	11	12	13	14	15	16	17	18	19	20
Recorded Value	9	9	10	12	13	14	18	19	24	26
$c_{i,L}$	8.5	8.5	9.5	11.5	12.5	13.5	17.5	18.5	23.5	25.5
$c_{i,U}$	9.5	9.5	10.5	12.5	13.5	14.5	18.5	19.5	24.5	26.5
Patient	21	22	23	24	25	26	27	28	29	30
Recorded Value	29	31	42	45	50	57	60	71	85	91
$c_{i,L}$	28.5	31	41.5	45	50	56.5	59.5	71	85	90.5
$c_{i,U}$	29.5	∞	42.5	∞	∞	57.5	60.5	∞	∞	91.5

Table 3.1: Data on leukemia remission.

3.2 Truncation

A careful reading of the previous section will show that censoring is really a property of the process of observation, not a specified statistical model *per se*. In contrast, what we usually mean by truncation relates to the specification

of distributions in models. There are two major types of truncation, *re-normalization* and *Winsorization*.

3.2.1 Re-Normalization

It may occur that, for whatever reasons, we would like to model certain random variables $\{Y_i : \imath = 1, \ldots, n\}$ with given distributions having densities $f_i(y|\boldsymbol{\theta})$ with common support $y \in \Omega$. The densities are indexed by i here to allow for nonidentical situations such as those involving covariates. But it may also occur that the support of these densities extends beyond the range of possible values for our random variables. A simple example of this would be in a problem for which we might like to use normal distributions for the error terms of an additive error regression, but the responses must be positive and there are observations near the origin. In such situations one option is to truncate the support of the assigned densities, and then re-normalize to maintain a proper model.

Consider a set of densities $\{f_i(y|\boldsymbol{\theta}) : i = 1, \ldots, n\}$ with common support Ω. Consider also a set of random variables $\{Y_i : i = 1, \ldots, n\}$ such that the possible values of these variables are contained in some connected subset $\Omega_0 \subset \Omega$. Almost always, Ω is some interval on the real line (including unbounded intervals in one or both directions) and Ω_0 is an interval that has upper and/or lower endpoints of smaller absolute value than those of Ω. Then we can assign the random variables distributions having densities

$$g_i(y|\boldsymbol{\theta}) = \frac{f_i(y|\boldsymbol{\theta})}{\int_{\Omega_0} f_i(t|\boldsymbol{\theta})\, dt} \tag{3.4}$$

with common support $y \in \Omega_0$.

Example 3.2

Consider a set of random variables $\{Y_i : i = 1, \ldots, n\}$ such that possible values are strictly positive but otherwise a simple linear regression might be appropriate. How might we model such a situation? One thought would be to take the error terms in a simple linear regression to have normal distributions truncated below at zero. This model would be, for $i = 1, \ldots, n$,

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

where the ϵ_i will be taken as independent and identically distributed with probability density function

$$f(\epsilon_i|\sigma^2) = (1/2)\,(2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2}\epsilon_i^2\right); \;\; \epsilon_i > 0. \tag{3.5}$$

This might be quite a difficult model to interpret, since we have what is essentially a simple linear regression but with only positive "error terms". That is, deviations from the regression line are not deviations from expected values because the expected values of the Y_i are no longer given by the regression line (and the variances are no longer given by σ^2). A better option might be to forego an attempt to write the model as an expression for the Y_i (as $Y_i =$ something), and simply assign these variables normal distributions truncated below at zero and having parameters $\beta_0 + \beta_1 x_i$ and σ^2. That is, assuming covariates $x_i > 0$ have a positive influence on the expected values of the responses, the model consists of assigning Y_i the densities, for $i = 1, \ldots, n$, $0 < \beta_0 < \infty$, $0 < \beta_1 < \infty$, and $\sigma^2 > 0$,

$$f(y_i|\beta_0, \beta_1, \sigma^2) = \left[1 - \Phi\left(-\frac{\beta_0 + \beta_1 x_i}{\sigma}\right)\right]^{-1} (2\pi\sigma^2)^{-1/2} \exp\left[-\frac{1}{2\sigma^2}(y_i - \beta_0 - \beta_1 x_i)^2\right], \tag{3.6}$$

where $\Phi(\cdot)$ is the distribution function (cumulative density) of a standard normal random variable. Now model (3.6) does not have the expected values

and variances of the Y_i equal to $\beta_0 + \beta_1 x_i$ and σ^2 either, but they are close, especially for large values of $(\beta_0 + \beta_1 x_i)/\sigma$.

3.2.2 Winsorization

It is not uncommon for statisticians to use distributions with unbounded support to model quantities that definitely have values they cannot be larger (or perhaps smaller) than. The weight of a collection of hummingbirds might be modeled with either normal or gamma distributions, even though we know a hummingbird cannot weight 3 metric tons. Actually, even modelling the weight of elephants would face the same difficulty. We may model chemical concentrations using gamma, inverse gamma, or lognormal distributions that have support on the entire positive line, even though we know there is some physical limit that applies to the problem. Our presumption in formulating such models is that tail probabilities become negligible rapidly and that, past a certain point, the distribution has essentially no probability mass.

There are some cases in which the scenario of the previous paragraph applies but we wish to formulate a model that is more reflective of physical reality. It may be the case, for example, that we know the particular maximum (or minimum) value that is possible. In other cases we may not have a known maximum (minimum), but we can pick a value that is reasonably larger (smaller) than any physically possible maximum (minimum). Winsorization may sometimes be used to develop an appropriate model in these cases. Rather than re-scaling a distribution based on a fixed truncation point (which is re-normalization) we simple "stack up" all the probability past a fixed point at that point. In the case of discrete distributions this amounts to nothing more than adding up all the probability past a certain value and

assigning it directly to that value. In contrast, Winsorization of continuous distributions then leads to a mixed continuous/discrete density, in which there is positive probability that the variable assumes a value equal to the Winsorization point.

Example 1.10 – Figs and Wasps

Ecologists are interested in systems that have evolved to reflect seed-eating pollinator mutualism. In such systems, pollinating insects or birds receive benefit from the plant in the form of food or substrate for laying eggs and the plants receive the benefit of pollination, allowing reproduction. One example is provided by figs (family *Moraceae*) and the wasps that pollinate them (family *Agaonidae*). Figs produce enclosed inflorescences (called syconia) that contain a large number of female florets (ovules – think of tiny enclosures in the shape of a cup) that produce chemicals that attract certain species of wasps. Foundress wasps enter these inflorescences and pollinate flowers while laying eggs into some of the ovules before dying within the inflorescence (Janzen, 1979). Wasp larvae develop for several weeks before males (who are wingless) emerge and compete for access to females. After mating, females collect pollen and exit the syconia through holes chewed out by male wasps. There are a fairly large number (hundreds) of ovules available in a fig syconia and yet this is still a finite resource available to the wasps. While the fig-wasp system can become quite complex (involving also a number of species of parasite wasps that gain benefit from the figs without providing any benefit to the plant), one of many questions that ecologists ask about this system is the amount of the available resource that is utilized by wasps. Data to address this question are gathered by collecting fig syconia, dissecting them

and counting the number of ovules that are occupied and unoccupied by wasps. One model that could be used in this problem would be a binomial (y occupied out of n total ovules). But the number of ovules is large and one could also contemplate fitting a model with Poisson random variables to such data, except that the support of Poisson probability mass functions is unbounded. As yet another alternative, then, one could formulate a model based on Winsorized Poisson probability mass functions of the form, for $\lambda > 0$ and positive integer R,

$$\begin{aligned} f(y|\lambda, R) &= \frac{1}{y!}\lambda^y \exp(-\lambda) I(y < R) \\ &+ \left[1 - \sum_{x=0}^{R-1} \frac{1}{x!}\lambda^x \exp(-\lambda)\right] I(y = R); \quad y = 0, 1, \ldots, R \end{aligned} \tag{3.7}$$

Clearly, the behavior of the probability mass function in (3.7) depends on the relative magnitudes of λ and R. If R is not substantially larger than λ it can easily occur that $Pr(Y = R) > Pr(Y = R - 1)$ which is typically not the behavior we would wish to have reflected. Kaiser and Cressie (1997) formulate a spatial model with Winsorized Poisson distributions but, along the way, demonstrate that as long as R is about 3λ to 4λ a Winsorized Poisson distribution behaves much like a regular Poisson distribution.